

Three-Channel Equivalence: Construction, Intuition, and Formal Rewriting in One Neural System

> **Draft v0.1 (English)** | 2026-08-11
> Target: TMLR / COLM / Cognition (theory + empirical)
> Companion: P1 (representation + generalization theory); P2 (compilation method)
> Evidence: exp50 (three-channel consistency), exp41 (symbol permutation), exp80 (step-skipping), exp53 (convergence).
Single official run_exp judge.

Abstract

The century-old split in the foundations of mathematics – intuitionism (a mathematical object exists iff it can be constructed), structuralism (an object is its position in a structure), and formalism (mathematics is a sign game) – is mirrored in a single neural system as three execution channels for the same mathematical object. We show that, in a token-native representation, the *construction* channel (definition-driven unfolding), the *intuition* channel (compiled weight execution), and the *formal* channel (definition-driven rewriting) are pairwise semantically equivalent: on a 2000-digit carry chain all three produce identical results (1.000). We argue this is the neural implementation of Computational Trinitarianism (Curry–Howard: programs = proofs = types), with the crucial difference that the equivalence holds between *execution channels of one system* rather than between logics. We develop the epistemology: intuition is the reasoning fast path – hard to obtain (config-dependent epochs), but easy to generalize by skipping construction steps; symbol permutation invariance (0.987) and zero-decay step-skipping (2000 digits, radix 60, 100% anonymization) establish that the intuition channel carries structure, not symbol semantics. Stance: anti-Platonist – intuition is compiled construction, not innate.

1 Introduction

****The split.**** In the foundations of mathematics, three answers compete for the question "what is a mathematical object?" Brouwer's intuitionism: an object exists iff it can be *constructed*. Bourbaki's structuralism: an object is its *position in a structure*. Hilbert's formalism: mathematics is a *sign game*; an object is its role in a rewriting system. The dispute is usually presented as philosophical; in this paper we observe that it is also *architectural* – the three positions correspond to three ways a computing system can realize the same object.

****The unification.**** A single neural system over a token-native representation (P1) exhibits all three channels:
- *Construction* (intuitionism): unfold the definition chain, step by step – existence by construction;
- *Intuition* (structuralism): compiled weights position the object directly – existence by location in the learned structure;
- *Formal* (formalism): the definition-driven rewriting engine – existence by the sign-manipulation rules.

We show experimentally that the three channels are pairwise semantically equivalent on the same inputs (exp50: construction = formal exactly; intuition 1.000 on a 2000-digit carry chain). This is the neural implementation of Computational Trinitarianism – Curry–Howard's programs = proofs = types – but with a decisive difference: the equivalence is *within one system's execution channels*, not between distinct logical systems.

****Why this matters.**** If construction, intuition, and formal rewriting agree on the same object, then the philosophical positions are not alternatives to be argued about but *facets of a single realization*. The disagreement over "what an object is" becomes a disagreement over *which channel to look through* – and all channels answer the same way. The engineering content is the compilation relation: construction can be compiled into intuition (P2), and both agree with rewriting.

****Contributions.****

- ****C1 (Epistemology)** – Intuition is compiled construction. The intuition channel is not a third mysterious faculty; it is the construction process compiled into weights. Anti-Platonist: no innate structure, no a priori object.
- ****C2 (Empirics)** – Three-channel equivalence. Construction = formal exactly, and intuition 1.000 on the same inputs, including a 2000-digit carry chain (exp50).
- ****C3 (Empirics)** – Intuition carries structure, not symbols. Symbol permutation invariance (0.987, exp41) and zero-decay step-skipping (exp80) show the intuition channel realizes the structure of the objects, not their signifiers.
- ****C4 (Epistemology)** – Acquisition and step-skipping are decoupled. Intuition is hard to obtain (config-dependent epochs; 3 simple / 8 full, exp53) but easy to generalize by skipping construction steps – the two facts are not in tension.

2 Three Positions, Three Channels

Position	Object is	Access	Our channel	Execution
Intuitionism (Brouwer)	a construction	construct it	Construction	unfold definition chain, step by step
Structuralism (Bourbaki)	a position	locate it	Intuition	compiled weights place it directly
Formalism (Hilbert)	a sign game	play it	Formal	definition-driven rewriting

The three channels share a single definition source: the token-native representation (P1 §4). Construction and formal read the same definitions; intuition is a compiled image of the same definitions. Equivalence is therefore not a coincidence of three independent systems but a property of one system viewed through three channels.

****Computational Trinitarianism.**** Curry–Howard identifies programs, proofs, and types as the same thing under three presentations. Our claim is analogous: construction, intuition, and formal rewriting are the same computation under three presentations – *within one system*. The difference from Curry–Howard is the locus of equivalence: not "program = proof = type" as distinct entities, but "one object's three execution paths agree."

3 Intuition as Compiled Construction

****The claim.**** The intuition channel is the construction process compiled into weights. Training (P1 §5) supervises the model with judgment outcomes while *excluding answer traces*: the model cannot copy the unfolding; it must induce the computation from definitions and compile it. The compiled product is intuition – the reasoning fast path.

****Hard to obtain.**** Compilation is not free. Acquisition is config-dependent: 3 epochs for the simple suite, 8 for the full logic-and-arithmetic suite (exp53). Until compilation completes, the intuition channel does not generalize (exp53: epochs 1–3 at 0.1–24%). This is the **obtain** side of the decoupling.

****Easy to generalize.**** Once compiled, the intuition channel generalizes by **skipping construction steps**: it jumps from input to output over the intermediate unfolding. Zero decay out to 2000 digits, radix 60, and 100% digit anonymization (exp80). This is the **generalize** side.

****Decoupling (C4).**** The two facts – hard to obtain, easy to skip – are not in tension. Obtaining requires compiling the structure (training); generalizing uses the compiled structure (inference). The theory of P1 explains both: definition completeness + relational sufficiency determine whether compilation succeeds; step-skipping is what compiled structure does on new inputs.

4 Experiments

All via the official run_exp judge path (full-sequence reconstruction). Model: exp02_supervised_s2 (2-layer, dim-64, <1 MB, judgment 1.000 on the full suite).

4.1 Three-channel equivalence (C2)

Input	Construction	Formal	Intuition
12+34	✓	✓	✓
123456789+987654321	✓	✓	✓
999...9 (20 digits)+1	✓	✓	✓
999...9 (2000 digits)+1	✓	✓	**1.000**

Construction = formal exactly on full output sequences (2/9/20/2000 digits). Intuition scores 1.000 on the same inputs. The three channels realize the same computation.

4.2 Intuition carries structure, not symbols (C3)

- ****Symbol permutation (exp41)****: 26080 digit renamings during training, tested on original symbols – invariance 0.996–0.987. The intuition channel realizes structural relations (successor/iteration/carry), not digit signifiers.
- ****Step-skipping zero decay (exp80)****: 2000 digits / radix 60 / 100% anonymization / joint, all 1.000. The compiled structure extrapolates without decay.
- ****Machine form (exp90)****: 20-digit addition compiles 82 construction tokens into 5 intuition tokens – the skip is visible in the token stream. Honest caveat: the atomic token is outside the model vocab (pad id 0); measurement is the sequence-length effect, a lower bound.

4.3 Acquisition–generalization decoupling (C4)

Epochs	1	2	3	5	8
Judgment acc (full suite)	0.001	0.214	0.244	0.781	0.996

Acquisition is monotone but config-dependent (8 epochs for the full suite). The same converged model then extrapolates with zero decay (§4.2). The "hard to obtain" and "easy to skip" facts coexist; they describe obtain (training) and generalize (inference), respectively.

5 Discussion

****What the equivalence does not say.**** We do not claim the three philosophical positions are identical in their epistemology; we claim they are equivalent as **execution channels of one system**. The historical disagreement concerned the nature of mathematical truth; our claim concerns the architecture of a computing system that realizes mathematical objects. The philosophical positions are models of what a mathematical object is; we show one system can realize all three models of the same object consistently.

****Anti-Platonism.**** Intuition is compiled construction, not an a priori faculty. If mathematical structure were innate (Platonism / nativism), then definitionally-impoorished concepts would still generalize; they do not (P1 §8: an impoverished vocabulary requires an order of magnitude more training; genuinely under-defined tokens never generalize). The intuition channel's competence is earned by compilation, not endowed.

****Positional vs constructive content.**** Structuralism says an object is its position; the intuition channel locates the object in the learned structure. Intuitionism says an object is its construction; the construction channel unfolds it. The equivalence of the channels shows the two descriptions coincide – for objects whose definitions are complete. This is the machine-level form of the old structuralist–intuitionist rapprochement.

6 Related Work

- ****Computational Trinitarianism**** (Curry–Howard): programs = proofs = types. We extend the three-way identification to execution channels of one neural system.
- ****Constructive mathematics / type theory**** (Martin–Löf 1984; Bishop): existence = constructibility. Our construction channel is its machine form.
- ****Structuralism / Bourbaki; Gärdenfors conceptual spaces****: object = position in a structure/geometry. Our intuition channel is its neural realization.
- ****Formalism / Hilbert; rewriting systems****: object = sign manipulation. Our formal channel is the definition-driven engine.
- ****Dual-process cognition**** (Kahneman): System 1/2. Our intuition/reasoning separation is the System 1/2 split with a compilation account of how System 1 is built.
- ****Dual-systems in math cognition**** (e.g. number sense / deliberate arithmetic): our three channels refine the two-system picture with a verifiable equivalence.

7 Limitations

1. The equivalence is demonstrated on the supported calculus (formalized definitions), not on all of mathematics.
2. The intuition channel is statistical, not guaranteed; equivalence is verified (held-out) and falls back to construction, not proven.
3. The atomization result (exp90) is a sequence-length lower bound, not a claim of learned atomic semantics.

8 Conclusion

The three positions of the foundations dispute – construction, structure, sign – are realized as three equivalent execution channels of one neural system. Construction = formal exactly; intuition agrees at 1.000; the intuition channel carries structure, not symbols; and it is hard to obtain but easy to generalize by skipping steps. Intuition is compiled construction. A century-old split in the foundations of mathematics is, at least for a formalized calculus, an architectural fact about how many ways one system can realize the same object – and all of them agree.

Appendix

- A. Full experiment tables: docs/paper_data/results.csv.
- B. Relation to P1 propositions (G0, G5, R) and P2 compilation.
- C. Formal-verification direction (Lean): construction $\equiv$ intuition $\equiv$ formal, for the supported calculus.